

Hand Gesture Controlled Robot

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Abstract

The aim of the project is to create a device that would help the physically disabled or the handicapped to use and maneuver objects conveniently in their everyday lives. As part of this project, a glove was created that consists of an MPU6050 accelerometer and an HC05 Bluetooth module that can control a 6DOF robotic arm. This robotic arm can be controlled to move in different directions and to pick and place objects with the aid of a glove that can communicate wirelessly with the robotic arm.

1. INTRODUCTION

1.1 Background

There are many people in this world who are unable to use their hands to their best potential, and this could be due to many different reasons. Some may have been handicapped at birth or may have developed certain medical conditions. This could also be due to accidents that may have affected the nerves in their hands, or it could simply be because of old age. Hence such handicapped individuals often have trouble carrying on with their daily lives as hands play a very important part in our daily activities.

1.2 Objectives

The objective of this project is to create a robot that would assist the physically disabled and the handicapped to carry on with their daily tasks more smoothly and conveniently. The robot must be able to move in different directions and to pick and place objects as determined by the user. This is done with the aid of a glove the user can wear on his hand which communicates wirelessly with the robot and controls it.

2. PROJECT DESCRIPTION

2.1 Physical Description of the Hand Gesture Controlled Robot

Robotic arm

Robotic arm is controlled by the servo motor which can move each joint and the gripper and the stepper motor for the base part of the robotic arm. The motors can be processed by the Arduino uno board and the driver with the power supplies maximum of 5V for all the 6 servo motors. To receive the data from the robotic glove, Bluetooth module is used for all the real time information.

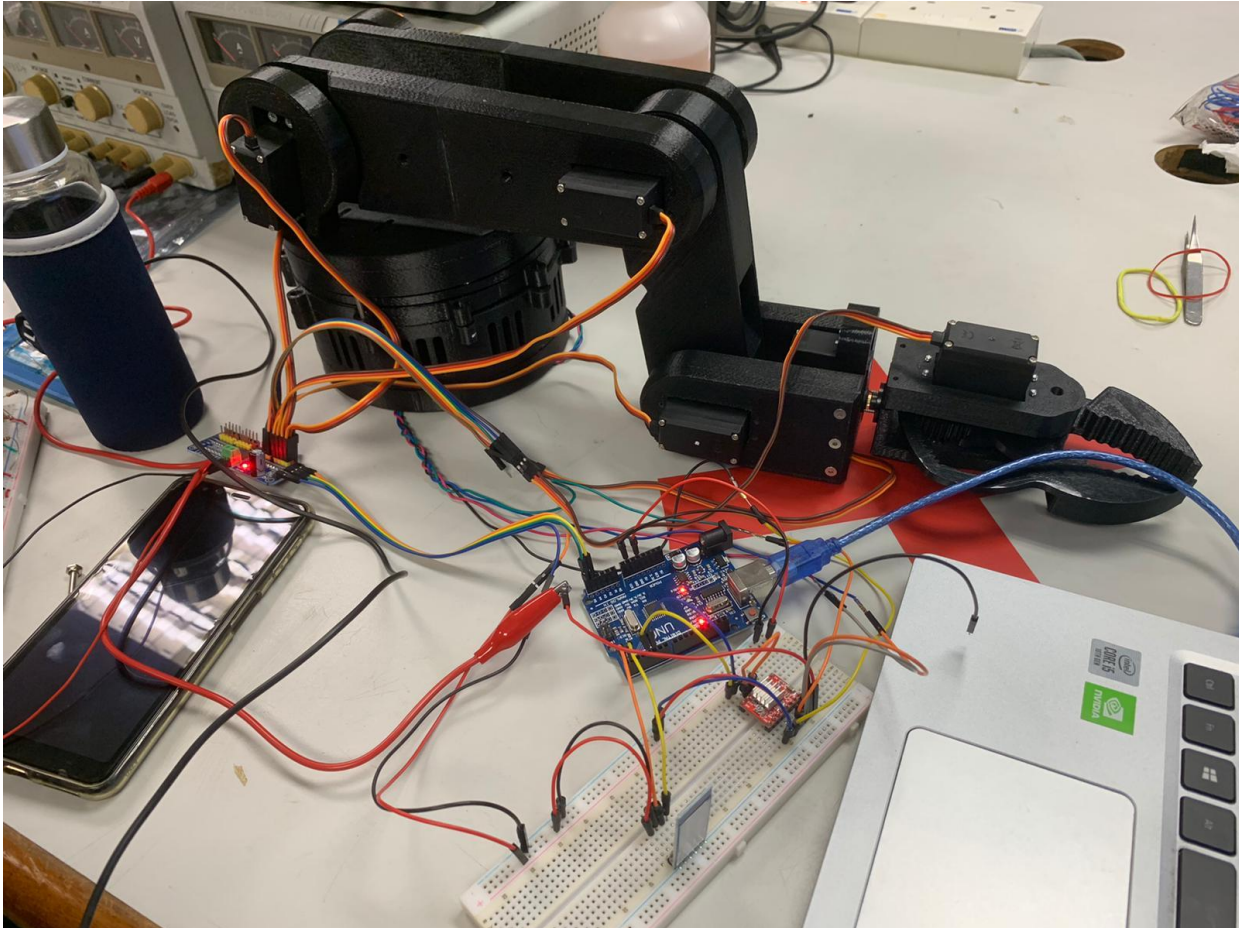


figure 1 - Assembled Robotic Arm

Haptic Glove

The robotic arm and the haptic glove are connected by a Bluetooth module, as described above. This allows it to send data from 3 flexible sensors and 2 accelerometers for the joints (up and down), the gripper (open and close) and the arm (left and right) respectively. It is important to have fixed and accurate hardware connections as the flex sensors and accelerators are quite sensitive.

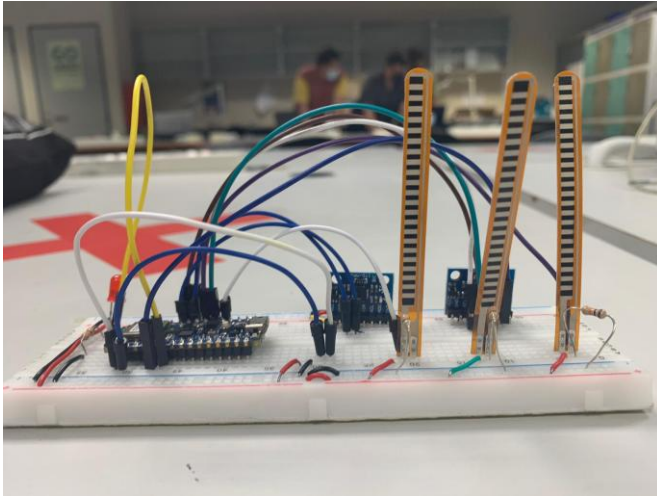


figure 2 - Circuit Patching of Haptic glove

2.2 Processing Circuit

Processing circuit consists of 1 stepper motor, 1 Bluetooth module, 6 servo motors, stepper motor driver, servo driver (PCA9685) and the Arduino uno according to the figure 3.

Haptic glove includes 3 flex sensors, 2 accelerators, Arduino nano, 1 Bluetooth module and 1 LED.

Robotic arm connection

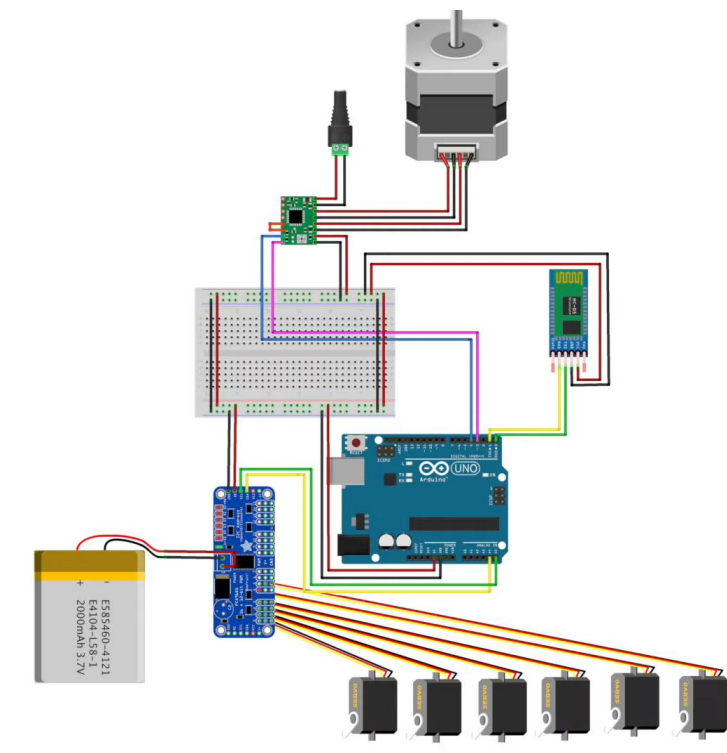


figure 3 - Schematic Diagram of the Robotic Arm Circuit

Haptic glove connection

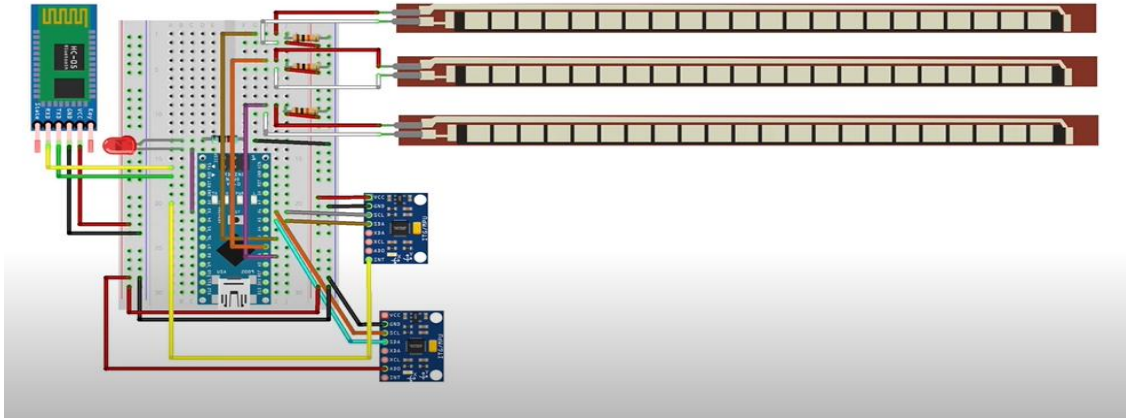


figure 4 - Schematic Diagram of the Haptic Glove

2.3 Process Description

For the processing, the robotic arm and the haptic glove are connected with the Bluetooth module. The haptic glove consists of 3 flex sensors and 2 accelerators which can transfer all the data via Bluetooth module that make the robotic arm to be move in every preferred direction and it's generated by the Arduino nano. After sending the data, the Bluetooth module from the robotic arm circuit received the data and the servo motors which are connected in every joint and the stepper motor for the base part are generated by the Arduino uno and the drivers.

3. PROJECT DEVELOPMENT

3.1 Research & Development

Some research on how the Arduino programming to control the robotic arm was done priorly. Coding is one of the main points of this project and thorough research is needed. Thorough research on coding is important so that no technical errors will be made during the final testing of this Project. Through researching from different resources such as websites and books, coding for combining the haptic glove and the robotic arm was finalized. The final project was settled down after two months of researching the hardware and coding of using this hand gesture robot. Several studies were previously done on the way Arduino is programmed to control the robotic arm. Programming is one of the main focuses of this project and needs to be thoroughly researched. Thorough research on coding was important so that no technical errors would occur in the final testing of this project. A final coding that combined the haptic glove with the robotic arm was determined by researching different resources such as websites

and books. Following two months of research into the hardware and coding using this hand-gesture robot, the final project was established.

3.2 Gathering & purchasing of components

Component's list has been completed and rechecked so that we don't need to waste time waiting for components in the next round. As for all the major components, such as servo motors and stepper motors, they were purchased online through Amazon. Pieces such as bearings and screws needed for each part on the robot arm were purchased from local online stores, such as RS Components. Upon arrival, the components are tested with Arduino test code to check that the components work.[2]

3.3 Fabrication of parts

The 3D model for the robotic arm was downloaded from an existing file from Thingiverse[3]. All the parts were 3D printed with the help of this Flash Print application and printed on a Dreamer printer at the Temasek Polytechnic Makerspace.

3.4 Assembly of Hardware Parts

The robotic arm was assembled individually by dividing it into three different parts, the base, the arm and the gripper. One by one, layer by layer, screws, bearings and metal balls were used to assemble these parts together to make the robot arm work successfully.

The Hardware Parts Used:

Robotic Arm

Servo Motor (MG966R) x 6

Servo Driver (PCA9685)

Battery Pack (5V, 2200 mAh)

Arduino Uno

Arduino HC-05

NEMA 17 stepper Motor

A4988 Stepper Motor Driver

12V Power supply

Haptic Glove

Flex Sensor x 3

MPU6050 (Accelerometer) x 2

Arduino Nano 33 ble

10k ohm Resistors x 3

220ohm Resistor

Builders Glove

9V Battery

9V Battery Clip

100nF Capacitor x 3

Arduino HC-05

3.5 Testing and Troubleshooting

After the servo motor and stepper motor are connected to the Arduino Uno, they are tested with test code to check if the components work successfully.

Robotic Arm Mapping

The table below shows the mapping of the values sent to the Robotic Arm through the glove.[1]

(P - Pinkie, T - Thumb, F - Index finger)

Number	Servo Name	Robotic Arm Action	Glove Sensor Values	Action
1	Base Joint	Rotate Base Left or Right	Acccelermotor_2 T, t	Move Elbow Left or Right

2 & 3	Shoulder Joint	Move Shoulder Joint Closer or Further	Pinkie Finger Flex – P, p	Bend or Extend Pinkie
4	Elbow Joint	Rotate Elbow Joint Closer or Further	Accelerometer_2 – C, c	Bend Elbow or Extend Elbow
5	Wrist Joint	Rotate Robotic Arm Joint Up or Down	Accelerometer_1 – U, D	Rotate Wrist Up or Down
6	Wrist Joint 2 (Rotate)	Rotate Robotic Arm (CW or CCW)	Accelerometer 1 – L, R	Rotate Wrist Left or Right
7	Gripper Joint	Close Claw Grip or Open Claw Grip	Index Finger Flex -F, f	Bend or Extend Index Finger

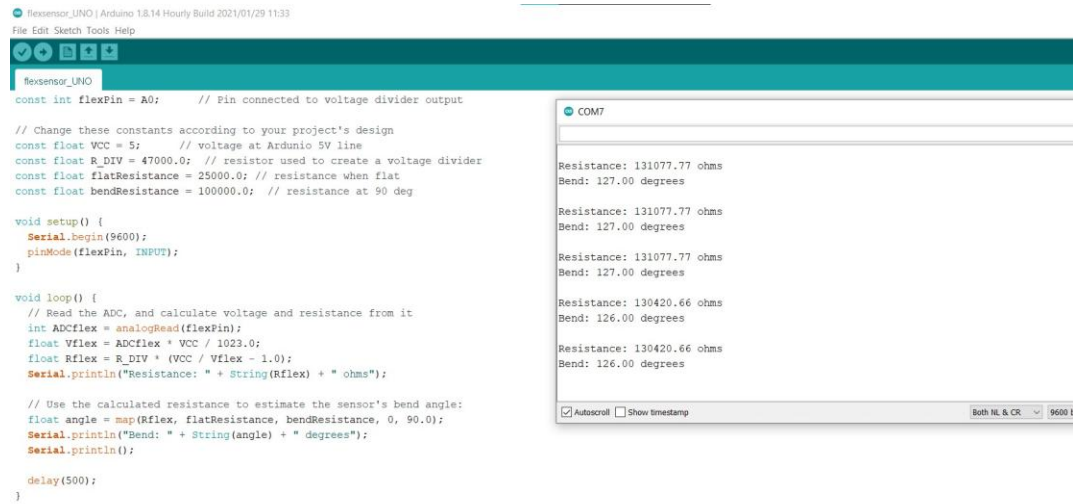
figure 5 - Mapping of the Robotic Arm

4. RESULTS

4.1 Test Results

With the connection of 6 servo motors and NEMA 17, the robot arm works properly with the help of Arduino code. Servo motors are able to work from 30 degrees to 180 degrees and the stepper motor in the base also rotates as specified in the code.

Nevertheless, the robot arm has difficulty in picking up and placing objects because some parts of the arm are heavy. The code was tested to solve this problem.



The screenshot shows the Arduino IDE interface. The sketch is named 'flexsensor_UNO'. The code defines a flex pin (A0) and constants for VCC, R_DIV, flatResistance, and bendResistance. The setup function initializes serial communication and pin mode. The loop function reads the ADC value, calculates resistance, and estimates the bend angle. The serial monitor shows the following output:

```
Resistance: 131077.77 ohms  
Bend: 127.00 degrees  
  
Resistance: 131077.77 ohms  
Bend: 127.00 degrees  
  
Resistance: 131077.77 ohms  
Bend: 127.00 degrees  
  
Resistance: 130420.66 ohms  
Bend: 126.00 degrees  
  
Resistance: 130420.66 ohms  
Bend: 126.00 degrees
```

figure 6 - Testing flex sensor values

5. CONCLUSION

The project group successfully developed the robotic arm controlled by hand gestures that can be used to help physically disable or handle the dangerous object to move conveniently. The degree of freedom of the robotic arm is successfully processed without any disturbance in rotating and the gripper part is working with perfect close and open for grabbing the object. However, the expected accuracy of the hand gesture is below the expected value.

To improve the accuracy, the prototype of the robotic glove is supposed to be printed the designed which could keep the circuit of the flex sensors and the accelerators in a fixed and tighter position so that the sender will send the more accurate glove data to the receiver as the flex sensors and the accelerators are not shaking while using the glove.

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Other Sources

- [1] Eben Kouao, DIY Arduino Robot Arm - Controlled by Hand Gestures | Full Tutorial, Jan. 11, 2021. [Video]. Accessed: Jul. 15, 2021. Available: <https://www.youtube.com/watch?v=F0ZvF-FbCr0&t=2s>

Appendices

- Appendix A: Assembled Robotic Arm
- Appendix B: Circuit Patching of Haptic Glove
- Appendix C: Schematic Diagram of the Robotic Arm Circuit
- Appendix D: Schematic Diagram of the Haptic Glove
- Appendix E: Mapping of the Robotic Arm
- Appendix F: Testing flex sensor values.